

$$1. x(t) = 1 + j e^{j2\omega_0 t} - j e^{-j2\omega_0 t} + 2e^{j5\omega_0 t} + 2e^{-j5\omega_0 t}$$

$$= 1 - 2\sin 2\omega_0 t + 4\cos 5\omega_0 t$$

$$= 1 - 2\cos(2\omega_0 t - \frac{\pi}{2}) + 4\cos 5\omega_0 t$$

$$T=10 \Rightarrow \omega = \frac{2\pi}{T} = \frac{\pi}{5}$$

$$x(t) = 1 + 2\cos(\frac{2\pi}{5}t + \frac{\pi}{2}) + 4\cos 5\pi t$$

$$2. \omega_1 = \pi \quad T_1 = \frac{2\pi}{\pi} = 2 \quad \omega_2 = \frac{5\pi}{3} \quad T_2 = \frac{2\pi \times 3}{5\pi} = \frac{6}{5}$$

$$[2, \frac{6}{5}] = 6 \quad T_0 = 6 \quad \omega_0 = \frac{\pi}{3}$$

$$x(t) = 2 + 2\sin 3\omega_0 t + \cos 5\omega_0 t$$

$$= 2 - j e^{j3\omega_0 t} + j e^{-j3\omega_0 t} + \frac{1}{2} e^{j5\omega_0 t} + \frac{1}{2} e^{-j5\omega_0 t}$$

$$a_0 = 2 \quad a_3 = -j \quad a_{-3} = j \quad a_5 = \frac{1}{2} \quad a_{-5} = \frac{1}{2}$$

$$3. |\cos \theta| = a_0 + \sum a_n \cos 2n\theta$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} |\cos \theta| d\theta = \frac{2}{\pi} \sin \theta \Big|_0^{\frac{\pi}{2}} = \frac{2}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} |\cos \theta| \cos 2n\theta d\theta$$

$$a_n = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} 2\cos \theta \cdot \cos 2n\theta d\theta - \frac{1}{\pi} \int_{\frac{\pi}{2}}^{\pi} 2\cos \theta \cos 2n\theta d\theta$$

$$2\cos \theta \cdot \cos 2n\theta = [\cos(2n+1)\theta + \cos(2n-1)\theta]$$

$$\int 2\cos \theta \cdot \cos 2n\theta = \frac{1}{2n+1} \sin(2n+1)\theta + \frac{1}{2n-1} \sin(2n-1)\theta + C$$

$$\int_0^{\frac{\pi}{2}} 2\cos \theta \cos 2n\theta d\theta = \frac{\sin(2n+1)\frac{\pi}{2}}{2n+1} + \frac{\sin(2n-1)\frac{\pi}{2}}{2n-1}$$

$$\int_{\frac{\pi}{2}}^{\pi} 2\cos \theta \cdot \cos 2n\theta d\theta = -\frac{\sin(2n+1)\frac{\pi}{2}}{2n+1} - \frac{\sin(2n-1)\frac{\pi}{2}}{2n-1}$$

$$a_n = \frac{2}{\pi} \left(\frac{\sin(2n+1)\frac{\pi}{2}}{2n+1} + \frac{\sin(2n-1)\frac{\pi}{2}}{2n-1} \right)$$

$$= \frac{2}{\pi} \frac{-2(-1)^n}{4n^2-1} = -\frac{4}{\pi} \frac{(-1)^n}{4n^2-1}$$

$$x(t) = \frac{2}{\pi} + \cos\left(\frac{2\pi}{T}t\right) - \frac{4}{\pi} \sum \frac{(-1)^n}{4n^2-1} \cos\left(\frac{4n\pi}{T}\right)t$$

$$a_0 = \frac{2}{\pi} \quad a_1 = 1 \quad a_k = -\frac{4}{\pi} \frac{(-1)^n}{4n^2-1} \quad a_k = 0$$

$$4. x(t) = \sum \delta(t-2n) - 2\sum \delta(t-2n+1)$$

$$T_0 = 2 \quad \omega_0 = \frac{2\pi}{2} = \pi$$

$$x(t) = \sum a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} \int x(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{2} [e^{-jk\pi \cdot 0} - 2e^{-jk\pi \cdot 1}]$$

$$= \frac{1}{2} (1 - 2\cos k\pi)$$

$$a_k = \frac{1}{2} [1 - 2(-1)^n]$$